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CHAPTER 1

PROBLEM DOMAIN

In today's world, the demand for electrical energy is increasing rapidly due to population growth, urban development, and the rise of electronic devices. Traditional power generation methods rely heavily on non-renewable resources such as coal, oil, and natural gas. These resources are gradually depleting and cause significant environmental pollution, leading to climate change and ecological imbalance. Therefore, there is a need for innovative, eco-friendly, and cost-effective methods of generating electricity.

At the same time, crowded public places such as railway stations, bus stands, malls, schools, airports, and parks experience continuous human movement. A large amount of mechanical energy in the form of footstep pressure is wasted daily without being utilized. This unused mechanical energy has the potential to be converted into electrical energy through piezoelectric technology.

However, the challenge lies in efficiently capturing this small, irregular, and low-voltage energy, storing it, and monitoring it in real time. Without proper conditioning, the generated voltage is unstable and cannot be directly used to power devices or be displayed for analysis.

To address these challenges, an IoT-enabled footstep energy generator is proposed. It focuses on converting footstep pressure into electrical energy using piezoelectric sensors, rectifying and storing the generated voltage, counting footsteps, and displaying the values on an LCD. The IoT component allows real-time monitoring and data collection for smart-city applications.

This project lies in the domain of renewable energy harvesting, smart systems, and IoT-based monitoring, aiming to reduce dependence on non-renewable energy sources while making efficient use of everyday human activities.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Modern society depends heavily on electrical energy for almost every activity. As cities expand and technology becomes more essential in daily life, the pressure on existing power generation systems continues to increase. Most of the world still relies on non-renewable sources like coal and petroleum, which not only cause pollution but are also running out. Therefore, researchers across the world are actively exploring new ways of producing clean and sustainable energy.

One interesting idea that has received a lot of attention in recent years is harvesting energy from human footsteps. Every day, millions of people walk in public places such as railway stations, airports, shopping malls, schools, and offices. Huge amounts of mechanical energy are produced through these footsteps, but all of it goes to waste. The concept of converting this wasted energy into useful electricity is both innovative and environmentally friendly.

Our project—IoT-Enabled Smart Footstep Energy Harvester with Real-Time Monitoring and Analytics—belongs to this emerging field. Before building our device, we studied existing research, global innovations, commercial installations, and academic prototypes to understand strengths, weaknesses, and potential areas of improvement. This literature survey summarises these findings.

2.2 Piezoelectric Energy Harvesting: Background and Fundamentals

Piezoelectric materials generate electric charge when mechanical stress is applied. This phenomenon, known as the piezoelectric effect, has been widely used in sensors, microphones, lighters, and medical devices. In the context of energy harvesting, piezoelectric discs or plates act as miniature power generators. When stepped on, the applied force deforms the material, creating a small AC voltage.

Past studies reveal that the output of a single piezo disc is very small. However, by connecting multiple sensors in a series-parallel arrangement and designing a suitable mechanical structure, the total energy output can be increased. Researchers have identified the following advantages of

using piezoelectric sensors for energy generation:

- They do not require an external power supply
- They are compact and easy to install
- They work purely based on human movement
- They are suitable for low-power IoT systems and basic electronics

However, they also noted that efficient power conditioning is crucial for stable output, since the raw voltage fluctuates with pressure.

2.3 Global Research and Existing Footstep Energy Systems

2.3.1 Early Academic Prototypes

Initial academic projects involved simple setups where a few piezo discs were placed under lightweight platforms. These early experiments proved that footsteps can indeed generate electricity, although the quantity was very low. Researchers used basic circuits such as diode rectifiers and capacitors to store the generated energy. Most of these systems were small-scale and used mainly for demonstration purposes.

2.3.2 Development of Piezoelectric Floor Tiles

As the concept matured, more refined piezoelectric floor tiles were developed. These tiles included multiple piezo elements arranged under a strong platform to distribute force evenly. Studies demonstrated that energy output increases significantly when:

- The number of piezo sensors is increased
- The mechanical design ensures uniform pressure
- An efficient rectifier and storage circuit is used

These tiles were able to power small LEDs and sometimes charge capacitors or low-power batteries. Still, they were not suitable for heavy loads.

2.3.3 IoT-Integrated Footstep Systems

With the rise of IoT, several recent studies integrated microcontrollers and communication modules to measure and transmit data such as footstep count, voltage output, and total energy generated. These projects showed that adding IoT greatly enhances the usefulness of the system by enabling:

- Remote monitoring
- Real-time data analysis
- Integration with smart-city dashboards

However, many IoT-enabled systems were still prototypes and not fully optimized for practical deployment.

2.4 International Advancements: Japan's Smart Footstep Technology

Japan has been a global leader in implementing practical footstep energy systems. One of the most notable examples is the installation of piezoelectric flooring by JR East in crowded railway stations such as Shibuya and Tokyo Station. These installations demonstrated real-world application by:

- Generating electricity from thousands of commuter footsteps
- Powering nearby LED information panels
- Collecting footfall data for crowd management
- Raising public awareness about sustainable energy

Similarly, piezoelectric tiles have been placed in Japanese shopping malls and airports, where they power small digital displays and interactive advertisement boards. These deployments highlight that while energy output is modest, the technology is valuable when combined with data analytics and smart infrastructure.

2.5 Commercial Technologies: Pavegen and Global Trials

Beyond Japan, companies like Pavegen (UK) have installed kinetic tiles in airports, stadiums,

and educational campuses. These tiles generate electricity through a combination of electromagnetic induction and piezoelectric materials. They also focus heavily on collecting data from foot traffic, offering both energy generation and analytics.

Several universities in the United States and Europe have experimented with similar technologies, emphasizing:

- Smart buildings
- Event-based energy harvesting
- Environmental monitoring
- Low-power IoT deployments

Most of these projects conclude that footstep energy harvesting is not meant for high-power applications but is excellent for sensor-based and educational purposes.

2.6 Common Components Used in Previous Projects

Earlier projects commonly included:

- Piezoelectric discs (PZT/PVDF)
- Bridge rectifier using 1N4007 diodes
- Capacitor banks for smoothing
- Arduino, PIC, or ESP modules
- Basic mechanical tiles made of wood or acrylic
- LEDs or resistive loads

These systems worked, but they lacked several advanced features that are now expected from modern prototypes.

2.7 Drawbacks in Earlier Footstep Energy Generation Projects

Although earlier research successfully proved the concept, several limitations were repeatedly observed:

2.7.1 Low Power Output

Many prototypes used too few piezo sensors, resulting in very low voltage and current output. This made the systems impractical for powering real devices.

2.7.2 Poor Mechanical Pressure Distribution

Some tile designs did not apply sufficient or uniform pressure on the sensors, reducing efficiency.

2.7.3 Limited Data Monitoring

Earlier systems often lacked:

Real-time voltage display

Step counting

Data logging

IoT integration

This limited their usefulness for analysis or practical demonstrations.

2.7.4 Basic Circuit Design

Many old circuits were inefficient due to:

- Wrong diode selection
- High forward voltage drop
- Lack of voltage regulation
- No protective components

2.7.5 No Scalability or Real-World Testing

Most setups were tested only in labs. Very few projects attempted real-world installation or

durability testing.

2.8 Improvements Introduced in Our Project

Our minor project, IoT-Enabled Smart Footstep Energy Harvester with Real-Time Monitoring and Analytics, focuses on overcoming the weaknesses observed in previous works. The improvements make our design more efficient, measurable, interactive, and suitable for demonstration.

2.8.1 Multiple Piezo Sensors for Higher Output

We used multiple piezoelectric sensors arranged to ensure better voltage generation. This directly improves performance compared to earlier systems using only 1–2 discs.

2.8.2 More Efficient Power Conditioning Circuit

We implemented a proper power conditioning stage consisting of:

- Bridge rectifier (using 1N4007 diodes)
- Smoothing capacitor to stabilize voltage
- Voltage divider for safe measurement
- BC547 transistor where needed

This produces a more stable and measurable DC output than earlier simple circuits.

2.8.3 Real-Time Information Display on LCD

Our project includes a 16×2 LCD that displays:

- Footstep count
- Real-time voltage generated

This feature was missing in many earlier works and makes our system more user-friendly and informative.

2.8.4 Step Counting Mechanism

We added an algorithm to count each footstep detected through voltage spikes. This creates additional functionality, turning the system into both an energy harvester and a footfall monitoring device.

2.8.5 IoT Expandability

Although optional for the minor project, our design is compatible with IoT modules like:

- ESP8266
- MQTT dashboard
- Blynk or other cloud apps

This offers future scalability that older systems did not support.

2.8.6 Better Mechanical Tile Design

We improved the structure of the footstep tile to ensure that pressure is distributed more evenly across sensors. This enhances energy generation during each step.

2.8.7 Practical Demo-Oriented Design

Our project focuses not only on generating energy but also on displaying, analyzing, and making the data meaningful for users. This gives it more practical value than earlier theoretical designs.

2.9 Summary

This literature survey shows that footstep energy generation is a promising concept with increasing global interest. Projects in Japan, the UK, and academic institutions around the world highlight the growing relevance of piezoelectric energy harvesting. While earlier research proved that footsteps can generate electricity, these systems often suffered from low output, inefficient circuits, and limited monitoring capabilities.

Our project addresses these gaps by implementing multiple sensors, a stable conditioning circuit, real-time LCD monitoring, step counting, and IoT expandability. These improvements make our system a more complete, user-friendly, and practical prototype suitable for minor project requirements as well as future enhancements.

CHAPTER 3

MAJOR OBJECTIVE AND SCOPE OF PROJECT

3.1 Major Objective of the Project

The primary aim of this project is to design, develop, and evaluate a smart footstep energy harvesting system that not only generates renewable energy but also integrates monitoring and analytics for improved usability and performance. The detailed objectives are as follows:

3.1.1. To harvest electrical energy from human footsteps using piezoelectric technology

One of the core objectives is to capture the mechanical pressure produced while walking and convert it into electrical energy through piezoelectric sensors. Human footsteps produce small bursts of mechanical force that are typically wasted. This project aims to efficiently collect this pressure and transform it into meaningful electrical output, demonstrating the concept of micro-energy harvesting.

3.1.2 To design an efficient electrical energy conditioning and interface circuit

Raw energy produced by piezoelectric sensors is irregular AC voltage. Therefore, another major objective is to design a proper interface circuit that includes:

- A full-bridge rectifier for AC-to-DC conversion
- Smoothing capacitors to stabilize voltage
- Voltage dividers to safely measure the output
- A transistor interface or protection circuit to protect sensitive electronics

This ensures that the harvested energy is properly conditioned, measurable, and safe for connecting to microcontrollers.

3.1.3 To enable real-time monitoring of voltage and energy output

A key objective is to make the system interactive and user-friendly. The project integrates a 16×2 LCD display to show real-time data such as:

- Instant voltage generated per footstep
- Cumulative footstep count
- Basic energy analytics

This real-time visualization helps users immediately understand how much energy each step produces.

3.1.4 To implement footstep counting for analytics and smart monitoring

In addition to generating power, the system is designed to recognize and count each individual footstep. Every time pressure is applied, a voltage spike is detected by the Arduino. This spike is interpreted as a step, and the system increments the footstep count. This supports:

- Footfall analysis
- Smart monitoring in public spaces
- Crowd density estimation

This objective transforms the project into a dual-purpose system: energy harvester + footfall analytics tool.

3.1.5 To integrate IoT-based remote monitoring and data visualization

While the base model uses LCD for local display, the project is designed with the ability to integrate IoT modules such as ESP8266 or NodeMCU. This enables:

- Remote monitoring of voltage and footstep count
- Cloud-based analytics
- Smart-city dashboard integration
- Real-time alerts and data transmission over Wi-Fi

The goal is to make the system compatible with modern IoT ecosystems, even if full IoT integration is optional in the minor project phase.

3.1.6 To evaluate the feasibility of footstep energy harvesting in real-world environments

This project also aims to study the practicality of installing such tiles in high-traffic locations like malls, railway stations, schools, and parks. The evaluation includes:

- Output performance under different loads
- Mechanical stability
- Consistency of voltage generation
- Possibility of long-term deployment

3.1.7 To develop a low-cost, eco-friendly, and scalable prototype

The objective is to build a budget-friendly system using easily available components, making it accessible for academic and small-scale applications. The project contributes to sustainability by promoting renewable micro-energy harvesting and energy consciousness among users.

3.2 SCOPE OF THE PROJECT

The project scope defines the boundaries, applications, and potential future expansions of the footstep energy harvesting system. The expanded scope is as follows:

3.2.1 Scope in Renewable Micro-Energy Harvesting

The project is limited to capturing small amounts of energy generated by footsteps. It is not designed for large-scale electricity production but is aimed at demonstrating the potential of micro-energy harvesting technologies. The scope includes generating enough energy for:

- Small LEDs
- Sensors
- Microcontrollers

- IoT nodes
- Low-power displays

3.2.2 Real-Time Monitoring and Local Display Scope

The system provides real-time visibility of the energy being harvested using an LCD display. The scope covers:

- Displaying voltage output
- Showing live footstep count
- Demonstrating how energy varies based on pressure

This real-time interaction helps users understand the connection between human movement and energy generation.

3.2.3 Scope for IoT and Remote Analytics

While IoT is optional in the basic prototype, the project is fully designed to support:

- Wi-Fi-based remote data monitoring
- MQTT/HTTP dashboards
- Mobile app-based visualization
- Cloud-based storage for long-term analytics

This scope positions the project in the category of modern smart-city IoT systems.

3.2.4 Scope for Use in Smart-City Infrastructure

The system can be expanded into large-scale installations in:

- Railway stations
- Airports
- University corridors
- Shopping malls

- Public parks
- Bus stands
- Exhibition venues
- Stadiums

These locations produce thousands of footsteps daily, making them ideal for footfall analytics and low-power energy harvesting.

3.2.5 Educational and Research Scope

The project is highly suitable for academic use, as it demonstrates multiple engineering concepts such as:

- Renewable energy
- Sensor interfacing
- Power electronics
- Arduino programming
- IoT data handling
- Real-time monitoring
- It can be used in labs, workshops, exhibitions, and engineering demonstrations.

3.2.6 Environmental and Sustainability Scope

The system encourages the idea of capturing wasted mechanical energy and converting it into useful electricity. It promotes sustainability by:

- Reducing reliance on traditional energy sources
- Encouraging green awareness
- Supporting eco-friendly technology development

3.2.7 Scope for Future Enhancements and Scalability

The project has significant scope for future development, including:

- Adding more piezo sensors for higher energy output
- Using supercapacitors for better energy storage
- Implementing machine learning for intelligent analytics
- Adding mobile app notifications
- Creating full smart tiles with embedded IoT modules
- Powering small street lights or display units in public spaces

These possibilities make the system versatile and open for long-term research.

CHAPTER 4

PROBLEM ANALYSIS AND REQUIREMENT SPECIFICATION

4.1 Problem Analysis

The modern world is witnessing a rapid rise in energy consumption due to increasing population, urban development, digitalization, and continuous growth in industrial activities. Electricity has become an essential resource for every sector, including transportation, education, healthcare, communication, and residential services. However, traditional methods of electricity generation primarily depend on non-renewable energy sources such as coal, crude oil, and natural gas. These resources are limited in nature and are also major contributors to environmental pollution, greenhouse gas emissions, and climate change. As a result, there is a growing need for alternate, renewable, and sustainable energy solutions that can reduce the burden on conventional power systems.

One of the less explored yet promising solutions is the concept of harvesting energy from human footsteps. Whenever a person walks, runs, or applies pressure on the ground, a considerable amount of mechanical force is produced. This force is usually wasted and not converted into any useful form of energy. However, by utilizing piezoelectric materials, it is possible to convert this mechanical energy into electrical energy. Piezoelectric sensors produce electrical voltage when they experience mechanical pressure. In high-traffic public areas such as railway stations, metro platforms, airports, malls, schools, and campuses, thousands of footsteps occur every day. This presents a significant opportunity for micro energy harvesting.

Despite the potential, earlier attempts in footstep energy harvesting faced several limitations. Many existing prototypes used only one or two piezoelectric discs, which resulted in very low voltage and insufficient current. The mechanical design of earlier tiles was also inefficient, leading to uneven force distribution on the sensors. This reduced the overall energy output. Furthermore, most earlier projects did not include any form of data measurement, monitoring, or analysis. Without real-time visibility of voltage, energy generation, or number of footsteps, the user had no way of understanding how efficiently the system was working. This lack of analytics

also made it difficult to assess the feasibility of installing such systems in real-world environments.

Another major issue with previous systems was the absence of IoT integration. In today's world, real-time monitoring, cloud storage, and remote accessibility have become critical components of modern engineering systems. Without IoT support, earlier footstep harvesting systems were limited to only basic voltage generation and LED demonstrations. They lacked the ability to send data to dashboards, perform trend analysis, or assist in smart-city applications such as footfall monitoring, crowd management, or energy prediction.

Many earlier projects also used basic rectifier circuits with high voltage drops, poor smoothing capability, and unreliable output. This made the voltage unstable and unsuitable for accurate measurement. The absence of appropriate voltage dividers and protection circuits also posed a risk to microcontrollers such as Arduino. Similarly, the mechanical construction in older systems often used weak materials like cardboard or thin wooden sheets, making the setup unsuitable for sustained use.

Therefore, there exists a clear need for an improved, reliable, and more intelligent footstep energy harvester that can not only generate energy but also provide meaningful insights through real-time monitoring and analytics. It should be durable enough for demonstrations, accurate in voltage measurement, capable of counting footsteps, and flexible enough to integrate IoT capabilities. This analysis identifies the shortcomings of previous systems and sets the foundation for introducing a more refined and advanced solution.

4.2 Requirement Specification

Based on the above problem analysis, the proposed project must satisfy a set of essential requirements to ensure that the system is functional, efficient, safe, and scalable. These requirements may be divided into functional and non-functional specifications.

4.2.1 Functional Requirement

The core functionality of the system revolves around effectively harvesting energy, conditioning the output, measuring it accurately, and presenting the information to the user in real time.

The first requirement is that the system must successfully convert mechanical pressure from footsteps into electrical voltage using multiple piezoelectric sensors. These sensors must be arranged in such a way that they can capture maximum force from each footstep. Since piezoelectric output is in AC form, the system must include an AC-to-DC conversion mechanism using a bridge rectifier. The rectified output must then be smoothed using appropriate capacitors to avoid sudden voltage fluctuations.

Another important requirement is the measurement of voltage generated by each footstep. For this, the system must use an Arduino microcontroller capable of reading analog voltage levels. Since the output from piezo sensors can be unpredictably high, especially under heavy force, the system must include a voltage divider to protect the Arduino from receiving voltage above its safe limits. The Arduino must process the readings and identify voltage spikes to count footsteps accurately.

The system must also include a display unit—preferably a 16×2 LCD—to show real-time voltage generation and footstep count. This feature helps users understand how much energy they produce with each step and how many steps have been recorded so far. The display should refresh continuously and provide clear, readable output.

Another functional requirement is designing the system to be IoT-compatible. Even if the base prototype does not implement cloud connectivity, the architecture must support future integration with Wi-Fi modules like ESP8266 or NodeMCU. This allows the system to transmit data to cloud platforms and mobile dashboards for analytics and remote monitoring.

The system must also safely handle energy flow, preventing damage to sensitive components. Proper wiring, secure connections, a stable power conditioning system, and adequate insulation must be ensured. If required, the design may include transistors, resistors, and protective components that improve circuit stability and safety.

4.2.2 Non-Functional Requirements

In addition to core functionality, the system must satisfy several performance and quality standards to ensure long-term usability and reliability.

The first non-functional requirement is reliability. The system should consistently generate measurable voltage for every footstep applied. It should not fail unexpectedly or produce highly unstable readings. The components must be able to withstand repeated mechanical pressure and environmental conditions during demonstration.

Durability is another critical requirement. Since the prototype involves people stepping or pressing on the tile, the platform must be strong enough to handle repeated loads. The selected materials should prevent bending, cracking, or sensor displacement. The system should also be efficient, meaning it should minimize energy loss during conversion, rectification, and smoothing. The circuit should use components with low forward voltage drop and high sensitivity.

Another non-functional requirement is user-friendliness. The interface, especially the LCD display, must provide clear information without requiring technical expertise. The system should also be safe to handle, with no exposed wires or sharp edges.

Cost-effectiveness is an essential non-functional requirement for a minor project. Components should be affordable, locally available, and easy to replace. This ensures that the system is accessible for academic use and demonstrations.

The system must also offer portability. Since this is a prototype, the entire setup should be lightweight and easy to transport for presentations or practical examinations.

Finally, the system should be scalable. It should support future expansion by allowing more piezo sensors, bigger tiles, or IoT modules for cloud-based analytics. This ensures that the project can evolve into a larger research model or a real-world installation.

4.3 Summary

The analysis clearly shows that earlier footstep energy systems lacked efficiency, real-time monitoring, durable construction, and IoT integration. The proposed system addresses these gaps by providing a more reliable, informative, and scalable solution. The requirement specification lays out the exact expectations for building a robust and intelligent footstep energy harvesting system suitable for modern smart-city applications.

CHAPTER 5

HARDWARE/SOFTWARE PLATFORM ENVIRONMENT

The development of an IoT-enabled smart footstep energy harvester requires a well-structured platform environment that combines both hardware and software elements. The overall performance, accuracy, and reliability of the system depend heavily on how effectively these two environments interact and support each other. The hardware platform primarily deals with physical sensing, energy conversion, measurement, and display, whereas the software platform focuses on processing, logic implementation, control, and communication. Together, they create a comprehensive ecosystem that enables real-time energy monitoring, step analytics, and future IoT scalability.

5.1 Hardware Platform Environment

The hardware environment includes all physical components responsible for energy harvesting, conversion, conditioning, processing, and display. These elements must operate cohesively to ensure smooth and reliable functioning of the system.

The foundation of the hardware platform is the piezoelectric sensor module, which converts footstep pressure into electrical energy. These sensors are placed beneath a strong tile platform and generate alternating voltage every time pressure is applied. To ensure higher and more stable output, the system uses multiple piezoelectric discs arranged in a suitable configuration.

Since the voltage generated by piezo sensors is AC in nature, the hardware platform incorporates a rectification stage. A full-bridge rectifier using 1N4007 diodes converts the alternating voltage into DC. A filtering capacitor smoothes fluctuations in the rectified output, producing a more stable voltage level. This stabilized voltage is processed by a voltage divider, ensuring that the Arduino receives safe and measurable DC input through its analog pins.

The Arduino Uno microcontroller is the core processing unit of the hardware platform. It reads analog voltage values, detects sudden voltage spikes created by footsteps, counts individual footstep events, and updates the LCD display with real-time information. Arduino is chosen due to its simplicity, reliability, availability of libraries, and ease of interfacing with sensors, displays, and communication modules.

To help users visualize system performance instantly, a 16×2 LCD display is integrated into the hardware setup. This LCD shows real-time voltage generated by each footstep and the total number of footsteps detected. The display significantly enhances the user-friendliness and practicality of the system by providing immediate feedback.

The system is also designed with provisions for future IoT integration, enabling expansion through Wi-Fi modules such as ESP8266 or NodeMCU for wireless data transmission and cloud-based monitoring.

- Hardware Components (Key Points)
- Piezoelectric sensors (multiple discs)
- Full-bridge rectifier using 1N4007 diodes
- Smoothing capacitor for stable DC output
- Voltage divider for safe analog input
- Arduino Uno microcontroller
- 16×2 LCD display
- BC547 transistor / protective circuit components
- Durable mechanical platform or tile
- Reserved space for IoT modules (ESP8266/NodeMCU)

These components together form the backbone of the system's energy harvesting, measurement, and display functionality.

5.2 Software Platform Environment

The software platform complements the hardware by providing data processing, control logic, program execution, and user interaction capabilities. It allows the interpretation of electrical

signals generated from the sensors and converts them into meaningful information such as voltage readings and footstep counts.

The main software tool used is the Arduino IDE, which provides a simple and flexible interface for writing, compiling, and uploading code to the Arduino Uno board. The programming is done using a C/C++ based language that supports analog-digital conversion, real-time data processing, LCD interfacing, and logic-based decision-making.

The software platform is responsible for several critical operations including reading analog voltage values, identifying voltage spikes corresponding to footsteps, implementing step counting algorithms, and updating the LCD. It ensures that users receive accurate and clearly presented information about the energy generated.

To support the future expansion into IoT-based monitoring, the software environment can interface with tools such as MQTT dashboards, Blynk, ThingSpeak, and Firebase. These platforms enable cloud storage, graphical visualization, and trend analysis of energy and footstep data.

- Software Components (Key Points)
- Arduino IDE for code development
- C/C++ based Arduino programming language
- Libraries for LCD display and analog reading
- Logic for voltage measurement and step detection
- IoT tools for data uploading (MQTT, ThingSpeak, Blynk, Firebase – optional)
- Serial Monitor for debugging
- Embedded firmware for real-time operation

This software environment ensures that the hardware output is processed efficiently and displayed accurately while also providing scope for future smart analytics.

5.3 Integration of Hardware and Software Environment

The seamless integration of hardware and software platforms is essential for achieving accurate

and reliable system performance. The Arduino Uno acts as a bridge between both environments, converting analog electrical signals into digital information that can be displayed, analyzed, or transmitted.

- Effective integration ensures that:
- Piezo-generated voltage is correctly captured and processed
- Real-time values are updated promptly on the LCD
- Footsteps are detected accurately through voltage spikes
- User feedback is immediate and clear
- IoT data transmission becomes possible when modules are added
- Hardware safety is maintained through proper software logic

The integrated environment also prioritizes energy efficiency, system stability, and modularity. The modular architecture ensures the system can be expanded in the future without major structural changes, making it suitable for research and real-world pilot installations.

5.3 Conclusion

The Hardware/Software Platform Environment forms the technological foundation of the IoT-Enabled Smart Footstep Energy Harvester with Real-Time Monitoring and Analytics. The hardware platform manages energy harvesting, conditioning, and display, while the software platform handles processing, logic, control, and future IoT communication. Together, they create a complete and scalable solution capable of generating renewable energy, analyzing usage patterns, and supporting smart-city innovations. This environment makes the project suitable for academic demonstrations, research studies, and practical IoT applications.

CHAPTER 6

DETAILED DESIGN

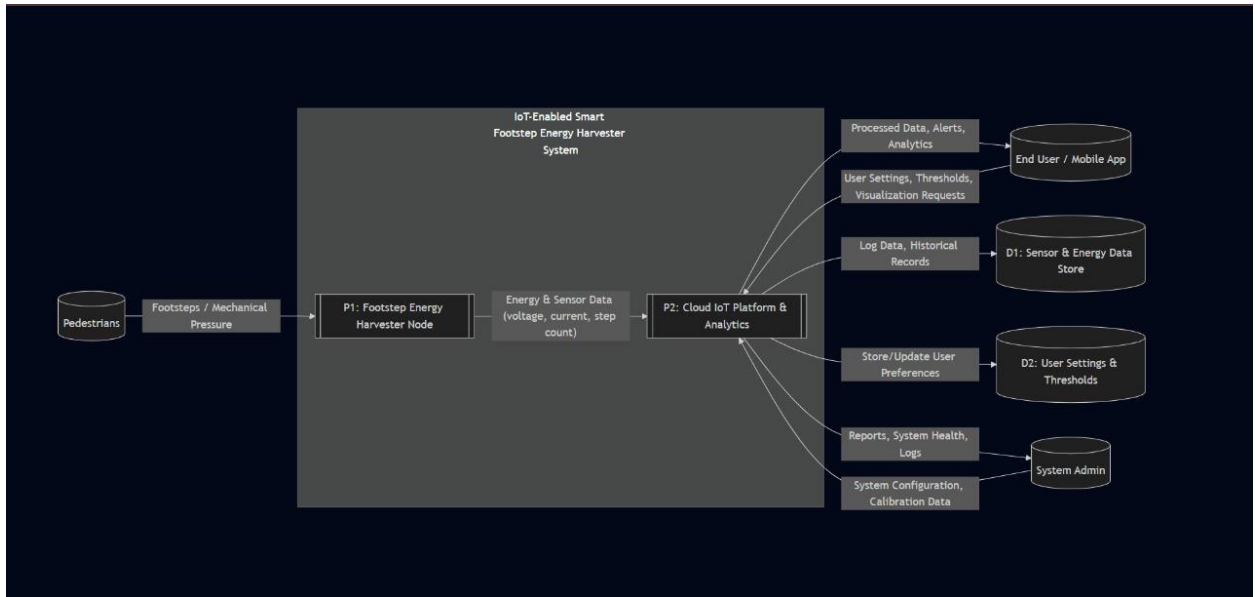


Fig. 6.1 0-Level DFD

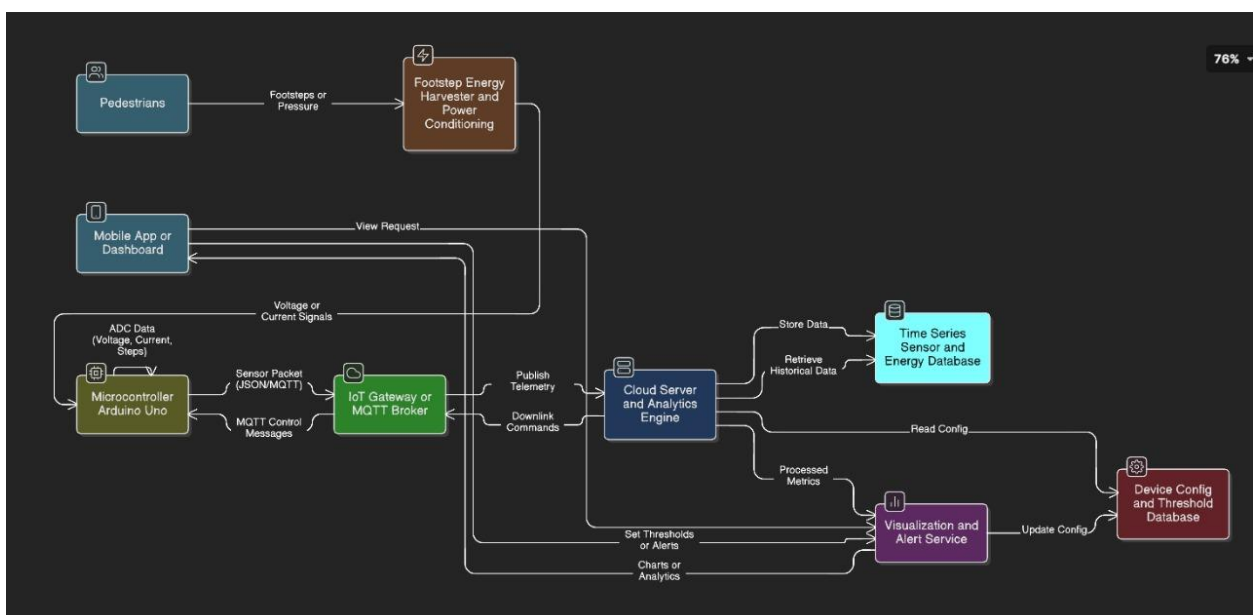


Fig. 6.2 1-Level DFD

CHAPTER 7

SNAPSHOTS



Fig 7.1

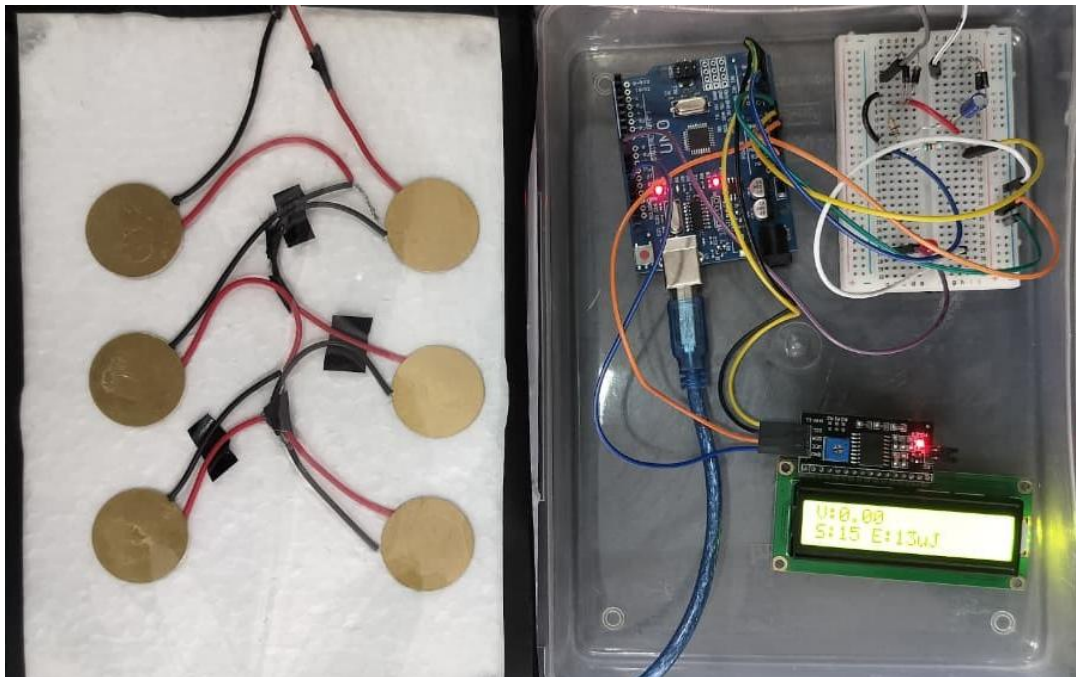
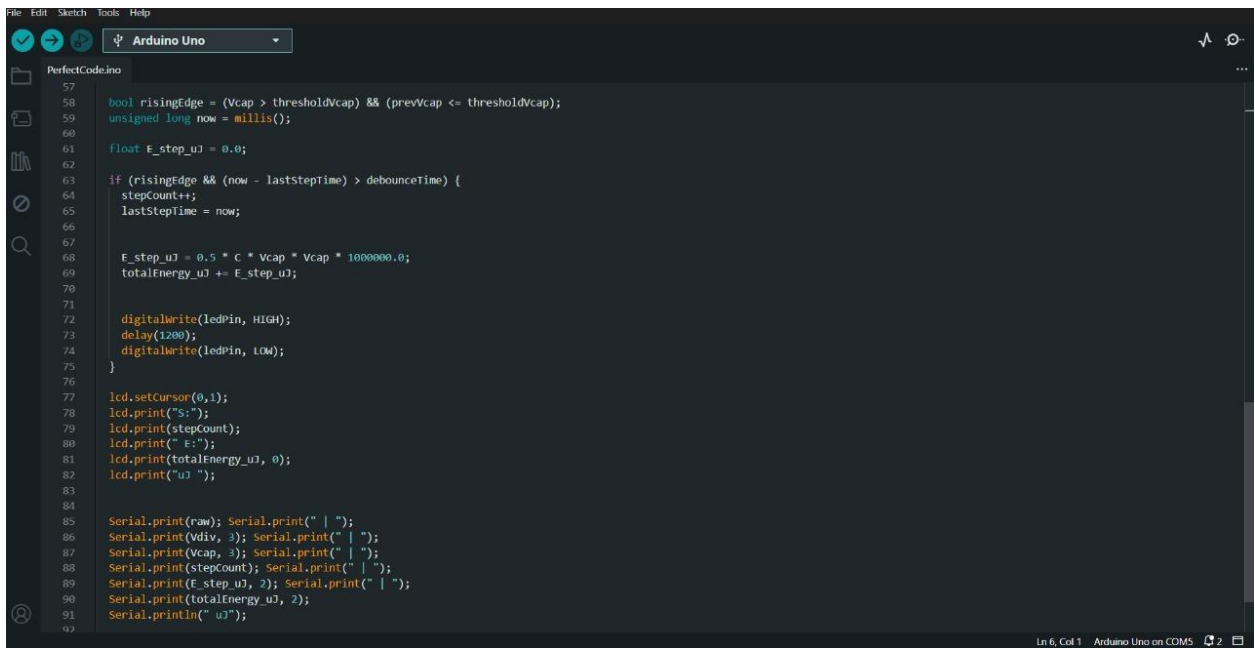


Fig. 7.2

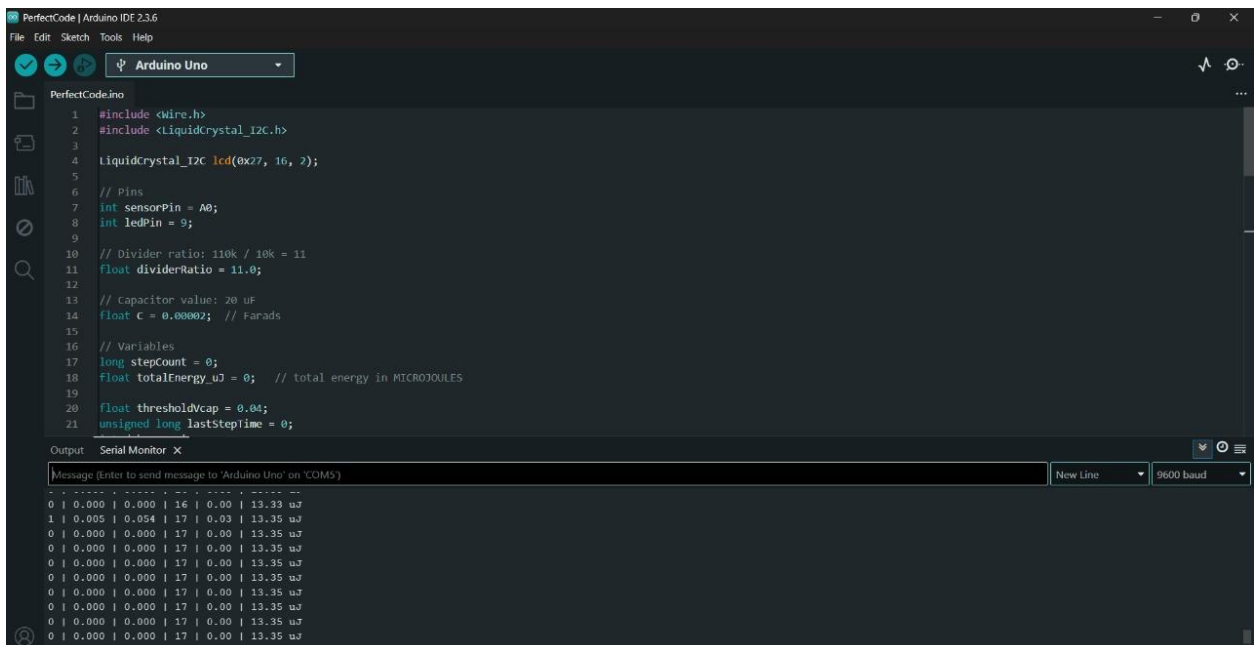
CHAPTER 8

CODING



```
57 bool risingEdge = (Vcap > thresholdVcap) && (prevVcap <= thresholdVcap);
58 unsigned long now = millis();
59
60 float E_step_uJ = 0.0;
61
62 if (risingEdge && (now - lastStepTime) > debounceTime) {
63   stepCount++;
64   lastStepTime = now;
65
66   E_step_uJ = 0.5 * C * Vcap * Vcap * 1000000.0;
67   totalEnergy_uJ += E_step_uJ;
68
69   digitalWrite(ledPin, HIGH);
70   delay(1200);
71   digitalWrite(ledPin, LOW);
72 }
73
74 lcd.setCursor(0,1);
75 lcd.print(":");
76 lcd.print(stepCount);
77 lcd.print(" E:");
78 lcd.print(totalEnergy_uJ, 0);
79 lcd.print("uJ ");
80
81 Serial.print(raw); Serial.print(" | ");
82 Serial.print(Vdiv, 3); Serial.print(" | ");
83 Serial.print(Vcap, 3); Serial.print(" | ");
84 Serial.print(stepCount); Serial.print(" | ");
85 Serial.print(E_step_uJ, 2); Serial.print(" | ");
86 Serial.print(totalEnergy_uJ, 2);
87 Serial.println(" uJ");
88
```

Fig. 8.1



```
1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3
4 LiquidCrystal_I2C lcd(0x27, 16, 2);
5
6 // Pins
7 int sensorPin = A0;
8 int ledPin = 9;
9
10 // Divider ratio: 110k / 10k = 11
11 float dividerRatio = 11.0;
12
13 // Capacitor value: 20 uF
14 float C = 0.00002; // Farads
15
16 // Variables
17 long stepCount = 0;
18 float totalEnergy_uJ = 0; // total energy in MICROJOULES
19
20 float thresholdVcap = 0.04;
21 unsigned long lastStepTime = 0;
```

Output Serial Monitor X

Message (Enter to send message to 'Arduino Uno' on 'COM5')

New Line 9600 baud

```
0 | 0.000 | 0.000 | 16 | 0.00 | 13.33 uJ
1 | 0.005 | 0.054 | 17 | 0.03 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
0 | 0.000 | 0.000 | 17 | 0.00 | 13.35 uJ
```

Fig. 8.2

CHAPTER 9

PROJECT LIMITATION AND FUTURE SCOPE

9.1 Project Limitation

Although the “IoT-Enabled Smart Footstep Energy Harvester with Real-Time Monitoring and Analytics” demonstrates the concept of energy harvesting very effectively, the system still faces several limitations. These limitations are mainly due to the physical nature of piezoelectric materials, prototype constraints, and the current level of electronic design implemented in this project.

One of the primary limitations is the low energy output produced by piezoelectric sensors. A single piezo disc generates a very small voltage and extremely low current. Even when multiple discs are connected, the total electrical output remains limited. As a result, the system is only suitable for powering small loads like LEDs, sensors, or microcontroller-level applications, and cannot support larger electrical devices.

Another limitation is the inconsistent and fluctuating voltage generated by footsteps. Since the piezoelectric output depends on the weight, pressure, and speed of the person stepping on the tile, voltage output varies from one footstep to another. This makes it difficult to maintain a stable energy supply without complex conditioning circuits.

The rectification and energy storage mechanism is another drawback. Using traditional 1N4007 diodes leads to voltage drop and energy loss. The capacitor used for storage has limited capacity, which means it cannot hold large amounts of charge. This restricts the system’s ability to store energy for extended usage.

Mechanically, the prototype design has limitations. The tile platform can withstand pressure for demonstration purposes, but it is not durable enough for long-term, high-footfall environments like airports or railway stations. A real-world installation would require waterproofing, strong

flooring material, better sensor housing, and robust insulation.

There are also limitations in the step detection algorithm. Footstep detection is based solely on voltage spikes, meaning sudden vibrations or accidental pressure may sometimes be counted as footsteps. This can lead to small inaccuracies in the step count.

In addition, while the system is IoT-ready, the base version does not include a fully integrated IoT module. Without adding ESP8266 or NodeMCU, the system cannot upload data to the cloud or provide remote monitoring capabilities.

Lastly, scalability and cost constraints exist. Adding more piezo sensors increases output but also raises the cost. Large-scale deployment would require careful planning, stronger materials, and more advanced circuitry, which may not be cost-effective initially.

Key Limitations :-

- Low energy output from piezoelectric discs
- Voltage inconsistency due to varying foot pressure
- Energy losses in rectifying diodes
- Limited energy storage capacity
- Prototype tile not suitable for heavy footfall areas
- Step detection may produce false triggers
- IoT module not included in the basic build
- Cost increases significantly with scaling
- Output not sufficient for medium/high power devices
- Sensitive wiring may need protection for long-term use

9.2 Future Scope

Despite its limitations, the project offers significant potential for improvement and wide-scale adoption in the future. The concept of harvesting energy from footsteps aligns with smart city development, sustainable energy initiatives, and IoT-based analytics applications. With further advancements, the system can be transformed into a highly efficient and commercially deployable solution.

A major area of future enhancement is the integration of IoT connectivity. Adding Wi-Fi-enabled modules like ESP8266, ESP32, or NodeMCU will allow real-time data transmission to cloud platforms. This would enable remote monitoring of voltage output, footfall count, and system health through mobile apps or dashboards. IoT integration can turn the system into a smart energy monitoring platform for public infrastructure.

The system has strong future scope in energy storage improvements. Incorporating supercapacitors, rechargeable lithium-ion cells, or advanced energy-harvesting ICs can significantly improve the ability to store and release energy. A hybrid system combining piezoelectric, solar, and electromagnetic harvesting will provide more stable and practical power output.

Another promising direction is mechanical optimization. Future versions may use spring-loaded mechanisms, flexible flooring materials, advanced pressure distribution plates, and stronger tiles to maximize force transfer to the sensors. Better housings and waterproof protections can make the system suitable for outdoor and high-footfall installations.

The system can also evolve into a powerful footfall analytics platform. With IoT and cloud connectivity, the system can record:

Daily footstep count

Peak traffic hours

Crowd movement patterns

Energy generation trends

Area-wise crowd distribution

These analytics are highly beneficial for malls, airports, metro stations, campuses, and smart city planners.

There is also strong scope for commercial deployment. With industrial-grade sensors and materials, the project can be installed in:

- Public pathways
- Corridors in schools and colleges
- Shopping malls
- Parks and recreational pathways
- Stadiums and event venues
- Railway and metro stations
- It can power LED signs, emergency lighting, small sensors, or Wi-Fi hotspots.

Additionally, the system can be enhanced with machine learning and AI. Using cloud-stored data, algorithms can predict crowd density, optimize energy harvesting locations, identify patterns, and help in smart urban management.

The environmental significance also contributes to its future potential. Harvesting renewable micro-energy from human footsteps supports sustainable development and promotes awareness of energy conservation among the public.

Future Scope

- Integration of IoT modules for online monitoring
- Cloud dashboards for footfall and energy analytics
- Use of supercapacitors or rechargeable batteries
- Improved mechanical tile with waterproofing and durability
- Large-scale deployment in public places
- Hybrid harvesting (piezo + solar + electromagnetic)
- AI/ML integration for predictive analytics
- Powering emergency lights, sensors, and small IoT devices
- Creating smart walkways and interactive public spaces
- Commercialization as a smart-city energy solution

9.3 Conclusion

The project successfully demonstrates the concept of energy harvesting and smart monitoring but also highlights areas that need improvement. With future upgrades in IoT, storage, design,

analytics, and scalability, the system has strong potential to become a valuable component of modern smart city infrastructure. It can evolve into a sustainable and intelligent platform that not only generates renewable electricity but also provides meaningful data for urban planning and crowd management.